Multimodal + Knowledge-Graph Document QA over Scientific Papers

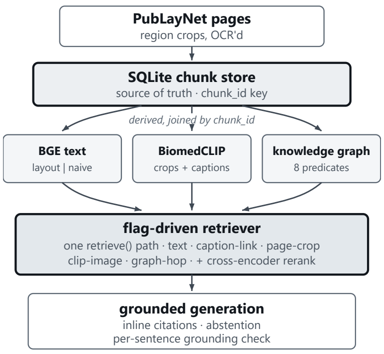
Does adding multimodal retrieval and a knowledge graph to a RAG pipeline actually beat a text-only baseline? This measures it on 500 scientific paper pages and 127 human-verified questions, and reports where it does not.

1. Problem & Approach

PubLayNet supplies region boxes and labels for scientific paper pages but no text, so every region is OCR'd from its crop. The HuggingFace parquet mirror used here drops the original filename, so PMC article IDs are unavailable in this copy and page = paper. The official PubLayNet distribution encodes the article in the COCO `file_name` field (`PMC<id>_<page>.jpg`), recoverable by joining the COCO `image_id` back to `val.json`. `page_id` and `paper_id` are stored separately so a real multi-page grouper swaps in behind `PaperGrouper` with no schema change — the KG null result below is therefore about this corpus copy, not about GraphRAG.

The design constraint is the fairness guarantee: **baseline and enhanced are one code path separated only by flags.** Dense BGE text retrieval always runs and that block alone *is* the baseline; each enhancement is an independently gated block that only adds to a candidate pool keyed by `chunk_id`. There is no `if baseline:` anywhere, so a difference in the numbers can only come from a flag, never from a divergent code path. A test asserts that the baseline and enhanced configs differ in nothing but their `flags` block, and that the 500-page config differs from the 50-page one only in which pages it ingests and where artifacts land.

2. System



Every arrow is config-selected: `AppConfig.from_yaml` resolves `extends:`, deep-merges so a child overrides only the keys it names, and validates strictly — an unknown or misspelled key is an error, not a silently ignored line; `describe()` prints the resolved settings at the top of every run.

So a report cannot disagree with what executed. Every external dependency sits behind an interface (`LLMClient`, `GraphBackend`, `VectorIndex`, `ChunkStore`, `RegionOCR`), and swapping the image encoder is a one-string change that also selects the index it reads.

Resolved for the reported run: BGE-base-en-v1.5 text, BiomedCLIP image, ms-marco-MiniLM cross-encoder, gpt-4o answering with gpt-4o-mini verification, `top_k` / `rerank_k` 10/10, `kg_hops` 2, `abstain_min_score` 0.45, CPU.

3. Evaluation & Results

Gold set. Every question is *selected* from a knowledge-graph edge — the answer and its supporting `chunk_id`s come from the edge, and the language model only phrases the selected fact. It never supplies an answer or evidence. Coverage is driven across six categories and all eight predicates. Three integrity measures matter:

- **Text-derived bias guard** — 23 questions written from a passage with the graph never consulted. If the graph-derived categories scored higher, the set would flatter the pipeline that built it. They score highest at baseline (0.83), which is the right direction.
- **Unanswerable items are verified absent**, not assumed: a candidate whose top cosine against the corpus clears 0.62 is rejected. At 500 pages this rejected **13 of 24** candidates that had been safely absent at 50 pages — without the check, those would have been silently answerable and the abstention result would have meant nothing.
- **`figure_value` answers are human-verified against the crop images.** Using the vision model to author the answer key for the category that measures the vision model would make any result circular.

127 questions: single_fact 56, text_derived 23, multi_hop 17, figure 12, unanswerable 11, figure_value 8. No category carries a low-sample flag — a first for this project.

Headline: the enhancement gap widens with corpus size.

	50 pages (n=55)		500 pages (n=127)	
	baseline	enhanced	baseline	enhanced
Recall@5	0.69	0.87	**0.57**	**0.91**
MRR	0.46	0.83	0.41	0.88
nDCG	0.52	0.84	0.47	0.89
Correctness	0.66	0.70	0.59	0.69
Faithfulness	0.73	0.78	0.82	0.82

Relative retrieval gain: **R@5 +26% → +60%; nDCG +62% → +89%.** The two ends move in opposite directions, and that is the result. Baseline degrades as the corpus grows — naive fixed-window chunking over ten times the candidates surfaces the right region less often. The enhanced configuration does not degrade; it improves. The usual pattern is the reverse, with small-corpus gains diluting as the corpus grows; here the 50-page numbers *understated* the enhancements.

Uncertainty, stated once. 95% bootstrap intervals over questions (10,000 resamples). Baseline R@5 **0.57 [0.49, 0.66]** → enhanced **0.91 [0.86, 0.96]**, non-overlapping. Correctness 0.59 [0.51, 0.67] → 0.69 [0.61, 0.77] overlaps marginally, but the *paired* difference is **+0.099 [+0.034, +0.168]**, excluding zero. Runs are single-seed by budget; run-to-run variance was measured rather than assumed — an identical configuration re-run over the same 17 questions moved correctness by **0.029**, so anything at or below ±0.03 is indistinguishable from re-running the same system.

Where the win comes from — before the table, not under it. It is layout chunking and caption anchoring, with reranking third. It is **not** image embeddings: `+clip`'s ΔR@5 is **+0.000 [+0.000, +0.000]**, exactly nothing. And on this corpus it is **not** the knowledge graph: ΔR@5 +0.026 [+0.000, +0.059] touches zero, and ΔCorrect +0.030 [+0.004, +0.064] has a magnitude that *is* the measured noise floor. The diagnosis matters more than the number — page = paper here, so every `multi_hop` question is answerable within one page, the regime where a graph helps least because both hops already sit in the same neighbourhood. A KG earns its cost when evidence is scattered across documents, which this corpus copy cannot supply. A null result about this corpus, not about GraphRAG.

Per-flag attribution (500 pages). One flag added at a time from a pure-text baseline.

config	R@5	MRR	nDCG	Correct	Decis
baseline	0.57	0.41	0.47	0.59	0.84
+layout	0.76	0.78	0.75	0.66	0.88
+clip	0.76	0.78	0.75	0.66	0.90
+caption	0.83	0.79	0.80	0.66	0.87
+kg	0.86	0.80	0.80	0.69	0.88
+rerank	**0.91**	**0.88**	**0.89**	**0.69**	0.89

Layout-aware chunking is the single biggest lever and alone recovers the entire baseline degradation. `+caption` is the figure route (figure R@5 0.50→1.00). `+kg`'s apparent correctness gain sits at the measured noise floor (above), and the feared dilution of `single_fact` by graph noise has now failed to appear on two corpora. `+rerank` is the largest retrieval step, a bigger contribution than at 50 pages — consistent with the rest, since reranking matters more when the pool is deeper. **`+clip` contributes nothing measurable**: identical retrieval to `+layout` on every metric.

`figure_value` and the visual path. This category was flat at 0.50 correctness across *every* config at 50 pages. At 500 pages with human-verified values it responds: **0.38 → 0.75 correctness, MRR 0.28 → 0.83** — though at n=8 those are [0.12, 0.75] → [0.38, 1.00] and overlap heavily, so the mechanism below is the evidence, not the jump. In a paired comparison where retrieval runs once and the answer is generated twice, the gold crop was retrieved **8/8** — but VQA did not beat text (strict 4/8 both; point-estimate 6/8 text vs 5/8 VQA). Exactly two questions differ and they cancel: on one, vision recovers a standard deviation that OCR linearisation lost; on the other, vision misreads a cell in the *correct* crop and cites it properly.

Operational, measured on the development machine (6 cores, 15.7 GB, CPU only; no API calls in the measurement). Retrieval-only query latency **p50 3.67 s, p95 13.44 s**, cold start 14.6 s, peak RSS **2.47 GB** with three transformer models and two FAISS indices resident. OCR ingest 119.7 min for 490 pages (14.7 s/page) and dominates corpus build; KG extraction 3.88 M tokens for $0.64. End-to-end latency including the answer model, and per-config API cost, are **not measured** — the harness never recorded per-question wall-clock or token counts, and obtaining them would cost budget. Total API cost for the full sequence: ~$5.8.

4. Findings & Limitations

Related work. These findings connect to established results. The compressed CLIP similarity distribution is the modality gap (Liang et al., 2022), known to widen under domain shift — which is why a domain encoder (BiomedCLIP) helps the score distribution; consistent with cross-modal alignment work, it does not by itself close the retrieval gap. Reaching a table crop through the prose of its page extends contextual retrieval (Anthropic, 2024), which reduces retrieval failures by prepending document context before embedding, to the cross-modal case: a crop, unretrievable by its own OCR, inherits retrievability from the text that contextualises it. The knowledge-graph component follows the GraphRAG paradigm (Edge et al., 2024); the coverage limitation observed at scale — a closed schema capturing only a fraction of freely-extracted relations on a topically broader corpus — is the documented schema-rigidity trade-off of predefined-schema GraphRAG. The retrieval-generation divergence, where reranking lifts multi-hop retrieval while depressing its correctness, is consistent with recent work arguing that multi-hop reasoning needs hop-separated rather than collapsed representations (Liu et al., 2025). Grounding and abstention follow the faithfulness objective of retrieval-augmented generation (Lewis et al., 2020). To our knowledge the multimodal-safety observation — that unconditional VQA converts a safe abstention into a confident, well-cited error unless gated on provenance — is not directly addressed in the existing literature, and we offer it as a contribution.

The CLIP modality gap is real and only half-fixable. General CLIP compressed 57 scientific crops into a 0.06-wide similarity band and separated rank 1 from rank 2 by 0.0003 — a coin toss (50-page corpus). BiomedCLIP widens the spread 6.7× and the top-1 margin 52×, and doubles image recall@1 (0.22→0.44). *Diagnosis:* that fixes the score *distribution*, not the retrieval *win* — recall@5 was unchanged, and at 500 pages `+clip` adds nothing on top of layout chunking. The caption anchor, not image similarity, is what makes figures retrievable.

A table cannot be retrieved by its own content. Transcribing tables and indexing the clean text made retrieval *worse* (gold table rank 31→40 and 19→36; cosine 0.536→0.459). *Diagnosis:* a table states values, not the concepts that make those values relevant — "lung cancer", "PAH" live in the page's prose, which outscores any possible table text. The fix is to reach the crop through its page (`use_page_crop_expansion`), which took gold-crop retrieval from 2/4 to 4/4 and holds at 8/8 on the larger corpus. Transcription was therefore **tested and excluded** as contributing nothing to retrieval.

Closed-schema GraphRAG trades coverage for tractability. Re-running schema discovery on the larger corpus: the closed 8 predicates cover only **22% of freely-extracted triples**. Two missing types recur — a predictor/association relation, and a "has measured value" relation. *Diagnosis:* the schema was consolidated from a toxicology/epidemiology slice and the 500-page corpus spans robotics, dentistry and policy editorials. A closed schema buys sparsity and precision (max entity degree 7, no hub explosion); an open one produced 46 distinct predicates over 30 chunks and never reused an edge type — a graph that cannot be traversed. The schema was kept unchanged and the gap reported. **This is also the structural reason `figure_value` finds no graph edges**: with no value-relation, `(mean age) —has_value→ (26.0 (8.6))` cannot be represented at all, which is a schema limit, not just an OCR one.

Breadth helps multi-hop and dilutes single-fact — twice, independently. The knowledge graph appears to lift multi-hop correctness, though the effect is at the noise floor, while adding context that a single-span question does not need. Chain-of-thought reproduces the same shape from the generation side: multi_hop 0.38 → 0.41 but single_fact (the control) 0.88 → **0.80**, with abstention rising in both subsets (0.24→0.29, 0.10→0.15). *Diagnosis:* reasoning surveys the retrieved set and folds in more that is topically related; on a question with one correct span, a broader answer is a worse one. **The multi-hop "gain" is not a result** — re-running the identical configuration moved the same 17 questions by +0.029, so the measured noise floor equals the entire effect. CoT is retained as an explainability feature, not an accuracy one.

Retrieval and answering come apart. Reranking lifts multi-hop R@5 0.62→0.82 while *dropping* its correctness 0.44→0.35. *Diagnosis:* the cross-encoder scores each chunk independently for query relevance, which is not the objective of assembling two chunks that must be combined. More starkly: figure R@5 goes 0.38→1.00 across the series while figure correctness goes 0.29→**0.25**. Solving retrieval did not solve answering, and reporting only the retrieval metric would have hidden that.

Attaching an image can convert a safe abstention into a confident error. Handed an off-topic crop, the vision model reads a real number off it and cites it. A score threshold provably cannot gate this: the wrong crop out-scored the right one (0.5534 vs 0.4589), and the inversion reproduces on raw OCR (0.5492 vs 0.5360) — both 50-page corpus. *Diagnosis:* a table's relevance is established by the page that discusses it, not by its own tokens — so the gate keys on **provenance**, with score demoted to a floor. *Limitation:* the guarantee weakens with corpus size, since with `top_k`=10 up to ten different papers can clear "the crop's page had a text hit", and one observed failure read the wrong paper's table.

Table values read out of OCR are wrong more often than right. Of 17 candidate values put to human verification against the crop images, **9 were rejected (53%; 8 of 14 among the OCR-derived proposals)** — misplacement, dropped exponents, label drift. *Diagnosis:* OCR has the right digits in the wrong structure and a vision transcription the right structure with some wrong digits, so neither may quote a value; the transcription is used only to find and score a table.

Abstention degrades at scale. Unanswerable abstain-rate falls from 1.00 at every config on 50 pages to 0.82 (0.91 at `+rerank`). *Diagnosis:* a larger corpus supplies more near-miss context, so a question the corpus cannot answer still retrieves material that looks answerable — and this is measured against a deliberately harder set, since the absence check had already removed the easy cases.

Extraction precision limits the graph. `occurs_in` correctness collapses to 0.31 (n=16) with retrieval fine at 0.81, and `treats` sits at 0.60 (n=15). *Diagnosis:* loose subject/object typing and observed direction inversions (`patients —treats→ chemotherapy`). Both are first candidates for an object-type constraint. Other standing limits: page = paper; title OCR 20.4% empty at scale; 76 of 410 crops have no caption region at all; the answer model is non-deterministic at temperature 0, so single-question comparisons are evidence about a mechanism, never fixtures.

5. Human-in-the-Loop & Explainability

Oversight is designed in layers rather than bolted on, because each layer catches a different failure. **Provenance** — every retrieved chunk carries how it was reached (`text`, `caption-link`, `page-crop`, `clip-image`, `graph-hop`), and every graph edge carries its source `chunk_id` and verbatim evidence sentence, so a graph-derived answer traces back to text. **Citation** — answers cite inline, validated against the context actually supplied. **Grounding** — each answer sentence is checked against its cited chunk, with an LLM verifier deciding only the borderline band. **Abstention** — a cheap code gate plus a prompt clause; the code gate is deliberately conservative and the semantic decision lives in the prompt.

The **review dashboard** turns those signals into a worst-first queue. Confidence is `0.5·Rn + 0.5·G` from retrieval strength and grounding — deliberately **label-free**, since building it from the gold-based retrieval metrics already in the logs would rank items by how right they are, which a deployed queue cannot know. Abstentions invert the priority: an abstention on weak retrieval is probably correct, while one on *strong* retrieval is the false-abstention failure mode and is what a reviewer should see first. Ranking blind, the queue surfaced the same defect the gold-backed metric measured — its top three items are figure questions, and figure decision-accuracy is 0.75.

Each item shows the answer, the three signals, provenance counts, every cited chunk with its source text pulled live from the store, the VQA gate's per-crop decisions, and the chain-of-thought trace where one exists. **The gold answer and the automated judge's verdict are hidden behind a toggle**, so the first judgement comes from the evidence rather than from the expected string — and revealing then audits both the reviewer and the judge. Verdicts (correct / incorrect / needs-review, plus a note) append to a log that records the confidence at review time, which is the raw material for asking whether confidence predicts human disagreement. The dashboard is strictly read-only over the locked results.

Honest limit: confidence is not a correctness estimate. Grounding scores a sentence against the whole cited chunk, so a short answer citing a long passage can score zero while being correct — observed in the queue, and the reason confidence is presented as a triage signal only.

References. Liang, W., Zhang, Y., Kwon, Y., Yeung, S., Zou, J. *Mind the Gap: Understanding the Modality Gap in Multi-modal Contrastive Representation Learning.* NeurIPS 2022. · Anthropic. *Introducing Contextual Retrieval.* 19 September 2024. · Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., Larson, J. *From Local to Global: A Graph RAG Approach to Query-Focused Summarization.* arXiv:2404.16130, 2024. · Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., Kiela, D. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* NeurIPS 33 (2020), 9459–9474. · Liu, J., Bai, J., Zeng, S. *Think Parallax: Solving Multi-Hop Problems via Multi-View Knowledge-Graph-Based Retrieval-Augmented Generation* (method: ParallaxRAG). arXiv:2510.15552, 2025.